



PTASs for secure dominating set in planar graphs and growth-bounded graphs

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ABSTRACT

Given a graph G with vertex set V and edge set E , a vertex set D is a dominating set (DS) of G , if every vertex $v \in V$ is either in D or adjacent to a vertex in D . If D is a DS of G and for every vertex $v \in V \setminus D$ there exists a vertex $u \in D$ adjacent to v such that $(D \setminus \{u\}) \cup \{v\}$ is also a DS of G , then D is a secure dominating set (SDS). The minimum secure dominating set problem (MinSDS) is to find an SDS of G with the minimum cardinality. In this paper, we design PTASs for MinSDS on planar graphs and growth-bounded graphs, that is, the computed solutions approximate the optimal value within factor $(1 + \varepsilon)$ for any constant $\varepsilon \in (0, 1)$.

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1. Introduction

Wireless sensor networks (WSNs) are widely used in various fields such as military, intelligent transportation, environmental monitoring, healthcare, etc [1,16,17,23,26]. There are abundant combinatorial optimization problems in WSNs. In this paper, we study one of them, the minimum secure dominating set problem.

Given a graph $G = (V, E)$ on vertex set V and edge set E , a vertex $v \in V$ is said to be *dominated* by a vertex set $D \subseteq V$ if either $v \in D$ or v is adjacent with a vertex in D . If D dominates all vertices in V , then D is called a *dominating set* (DS) of G . Vertex set D is a *secure dominating set* (SDS) if it is a DS of G and for any vertex $v \in V \setminus D$, there is a vertex $u \in D$ such that $(D \setminus \{u\}) \cup \{v\}$ is also a DS of G . Consider the application of arranging policemen at crossings of roads to monitor traffic. Every policeman can monitor the traffic on site and also traffics at its neighboring crossings. The problem of locating the minimum number of policemen to monitor all crossings is the *minimum dominating set problem* (MinDS). If an emergency occurs at some crossing v without policeman, a neighboring policeman, say originally located at u , can move to v to deal with the matter. It is desired that after the movement, all crossings are still monitored, that is, $(D \setminus \{u\}) \cup \{v\}$ is still a DS. Such a consideration leads to the *minimum secure dominating set problem* (MinSDS). Denote by $\gamma_{\text{SDS}}(G)$ the size of a minimum SDS of G , and call it the *secure domination number* of G .

There are many studies on SDS from algorithmic point of view. Only a few of them are on general graphs [4,8] and most of them are on special graphs [2,5,13,20,21,27]. In particular, Wang et al. [24] studied approximation algorithms for two variants of MinSDS, the *minimum secure connected dominating set problem* (MinSCDS) and the *minimum secure total dominating set problem* (MinSTDS). Linear time constant approximation algorithms were given for them on unit disk graphs and rectangle graphs. They also presented a PTAS (that is, a polynomial time $(1 + \varepsilon)$ -approximation algorithm) for

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MinSTDS on unit disk graphs and unit square graphs. This study motivates us to ask whether there is a PTAS for MinSDS on more general geometric graphs?

In this paper, we present PTASs for MinSDS on planar graphs and growth-bounded graphs. Planar graphs occur naturally in the real life, such as large scale integrated circuits, railway maps and chemical molecules. Many hard problems might have much better algorithms on planar graphs due to their sparsity and many interesting structural properties. The concept of growth-bounded graph was introduced by Nieberg et al. [19]. A graph G is said to be *growth-bounded* if there exists a polynomial bounding function $p(r)$ such that the number of independent vertices in r -hop neighborhood of v is no more than $p(r)$ for any $v \in V$ and $r \geq 0$. Roughly speaking, in a growth-bounded graph, the growth-rate of the number of independent objects is mild, can be controlled by a polynomial in terms of the size of the region. It captures an essential feature of many important classes of geometric graphs, including unit disk graphs and unit square graphs.

1.1. Related works

The concept of SDS was introduced by Cockayne et al. in [10]. There are many studies on SDS in special classes from graph theoretical point of view [9,12,22]. In this subsection, we only focus on those algorithmic results of MinSDS.

For the computational complexity of MinSDS, Merouane and Chellali [15] proved that computing the secure domination number is NP-hard, even when restricted to bipartite graphs and split graphs. Wang et al. [25] proved that for a general graph, MinSDS cannot be approximated within a factor of $(1 - \varepsilon) \ln |V|$ for any $\varepsilon > 0$, unless $NP \subseteq DTIME(|V|^{O(\log \log |V|)})$.

Only a few works study MinSDS on general graphs. Burdett and Haythorpe [4] proposed a binary programming formulation for MinSDS. Chen et al. [8] applied algorithmic game theory to study MinSDS. They presented a game framework for MinSDS, showed that every Nash equilibrium (NE) is a minimal SDS as well as a Pareto optimal solution; and designed a distributed local algorithm which converges to an NE in linear rounds of interactions.

There are many studies on variants of SDS. A *secure connected dominating set* (SCDS) of G , introduced by Cabaro et al. in [6], is a connected dominating set D_C of G (that is, a dominating set that induces a connected subgraph of G) such that for every vertex $v \in V \setminus D_C$, there exists a vertex $u \in D_C$ adjacent to v with $(D_C \setminus \{u\}) \cup \{v\}$ still being a connected dominating set. The *minimum secure connected dominating set problem* (MinSCDS) is to find an SCDS with minimum cardinality. Burdett and Haythorpe [4] provided a binary programming formulation for MinSCDS. A *secure total dominating set* (STDS) of G , introduced by Benecke et al. in [3], is a total dominating set D_T of G (that is, any vertex $v \in V$, not merely those vertices in $V \setminus D_T$, has a neighbor in D_T), such that for every $v \in V \setminus D_T$, there exists $u \in D_T$ adjacent to v with $(D_T \setminus \{u\}) \cup \{v\}$ still being a total dominating set. Finding an STDS with minimum cardinality is known as the *minimum secure total dominating set problem* (MinSTDS). Kumar and Reddy [14] designed a $(\Delta(G) + 1)$ -approximation algorithm for MinSCDS, where $\Delta(G)$ is the maximum degree of graph G . Wang et al. [24] designed constant approximation algorithms for MinSCDS and MinSTDS on unit disk graphs and rectangle graphs, as well as PTASs for MinSTDS in unit disk graphs and unit square graphs.

1.2. Our contribution

The contributions of this paper are summarized as follows.

(i) We propose a PTAS for MinSDS on planar graphs, that is, a polynomial time algorithm that computes a feasible solution which approximates an optimal solution within factor $1 + \varepsilon$. The algorithm employs a local search method inspired by Chan and Har-Peled [7] (when studying the maximum independent pseudo-disk problem) and Mustafa and Ray [18] (when studying the minimum hitting set problem with half-spaces in $\mathbb{R}^3$ and admissible regions in $\mathbb{R}^2$). To estimate the gap between a global optimal solution and the local optimal solution obtained by the local search method, a *separator theorem* was used on an auxiliary planar graph. A property that is crucial to the analysis is the *locality condition*, which guarantees that after a switch of objects, feasibility is kept. For the study of MinSDS, a challenge lies in how to construct an auxiliary planar graph to meet the locality condition. This is done in Section 2.

(ii) We design a PTAS for MinSDS on growth-bounded graphs. The algorithm divides the graph into small subgraphs with bounded radius, computes minimum local solutions for each subgraph, and then assembles them into a feasible solution to the original problem. One challenge is how to bound the error incurred by piecing up those subsolutions, which occurs at the boundaries of the subproblems. For this purpose, we obtain subproblems in an adaptively growing manner, so that the boundaries are “thin” enough not to incur too much error. A concept called 2-separated collection is used to help deriving a lower bound for the optimal value. Another challenge is: can we compute minimum local solutions to the subproblems in polynomial time? To answer this question, we show that due to growth-boundedness, each subproblem has its minimum local solution bounded by a constant, and thus an exhaustive search can find it in polynomial time. The algorithm and its theoretical analysis form the main part of Section 3.

Section 4 concludes the paper and discusses some future research directions.

2. PTAS for MinSDS in planar graphs

In this section, we present a PTAS for MinSDS on a planar graph, using a local search method. First we introduce some necessary symbols. Given a graph $G = (V, E)$, for $v, u \in V$, let $d_G(v, u)$ be the *distance from v to u in graph G* , that is, the length of a shortest path from v to u in G . For two vertex sets $X_1, X_2 \subseteq V$, let $d_G(X_1, X_2) = \min\{d_G(v, u) \mid v \in X_1, u \in X_2\}$.

For a vertex $v \in V$, denote its *open neighborhood* by $N_G(v) = \{u \in V \mid (u, v) \in E\}$ and its *close neighborhood* by $N_G[v] = N_G(v) \cup \{v\}$. For vertex set $X \subseteq V$, let $N_G(X) = (\bigcup_{v \in X} N_G(v)) \setminus X$ be the open neighborhood of X . When there is no danger of ambiguity, we shall omit the subscript G in the above notations. For an SDS D of G , a vertex $v \in V \setminus D$ is said to be *secured* by some vertex $u \in D$ if v is adjacent to u and $(D \setminus \{u\}) \cup \{v\}$ is also a dominating set of G . For a vertex in D , we regard it secured by itself.

2.1. The Algorithm

Given a planar graph $G = (V, E)$, a local search algorithm is given in Algorithm 1. It starts from the trivial feasible solution $D = V$, and recursively improves the solution by a *local switch*, that is, replacing a vertex set $Z_1 \subseteq D$ with size $|Z_1| \leq b$ by a vertex set $Z_2 \subseteq V \setminus D$ with size $|Z_2| \leq |Z_1| - 1$, keeping the feasibility of the solution, where $b = O(1/\varepsilon^2)$ is a constant related with the error term ε and the constants whose existence is guaranteed by Theorem 2.3. The algorithm terminates until no more local switch improvement is possible, at which time we say that a *local optimal solution* is obtained.

Algorithm 1 Local Search Algorithm for MinSDS

Input: A planar graph $G = (V, E)$ and a real constant $0 < \varepsilon < 1$.

Output: An SDS D of G .

- 1: $D \leftarrow V$; $b \leftarrow c \left(\frac{3(c_1+c_3)}{\varepsilon} \right)^2$, where c_1, c_2, c_3 are the constants in Theorem 2.3;
 - 2: **while** $\exists Z_1 \subseteq D$ with $|Z_1| \leq b$ and $Z_2 \subseteq V \setminus D$ with $|Z_2| < |Z_1|$ such that $(D \setminus Z_1) \cup Z_2$ is also an SDS of G **do**
 - 3: $D \leftarrow (D \setminus Z_1) \cup Z_2$;
 - 4: **end while**
 - 5: Output D .
-

Lemma 2.1. Algorithm 1 runs in time $|V|^{O(1/\varepsilon^2)}$.

Proof. Algorithm 1 starts from the trivial feasible solution with size $|V|$. In each iteration before termination, the size of the solution is strictly reduced. So the number of iterations in Algorithm 1 is at most $|V|$. For each iteration, checking whether the solution can be improved by switching vertex sets Z_1 and Z_2 with $|Z_1| \leq b$ and $|Z_2| < |Z_1|$ takes time $O(|V|^{2b})$. By the value of b set in the algorithm, the total running time is $|V|^{O(1/\varepsilon^2)}$. $\square$

2.2. Performance analysis

Suppose that D is the output of Algorithm 1. Denote by OPT a minimum SDS of G . For simplicity of statement, we assume that $D \cap OPT = \emptyset$. Such an assumption is without loss of generality. In fact, if $D \cap OPT \neq \emptyset$, then working on the vertex sets $D \setminus OPT$ and $OPT \setminus D$ instead of D and OPT , the following results still hold.

To estimate the size of D , we construct an auxiliary graph $G' = (V', E')$ as follows: $V' = D \cup OPT$; for any vertex $v \in V$, because both D and OPT are SDSs, there exist a vertex $u_1 \in D$ and a vertex $u_2 \in OPT$ such that v is secured by both u_1 and u_2 , we link u_1, u_2 in G' (vertex v is called a *witness* of edge $(u_1, u_2) \in G'$).

The next lemma shows that G' is a planar graph.

Lemma 2.2. For a planar graph G , the graph G' constructed as the above is also planar.

Proof. For an SDS D and a vertex $v \in V \setminus D$, denote $E_v^D = \{(v, u) \in E(G) \mid v \text{ is secured by } u \in D\}$. Since D is an SDS, any vertex $v \in V \setminus D$ has $E_v^D \neq \emptyset$.

Note that G' can also be constructed through the following operations:

Operation 1. Let $E_1 = \{(v, u) \in E(G) \mid \{v, u\} \subseteq D \text{ or } \{v, u\} \subseteq OPT \text{ or } \{v, u\} \subseteq V \setminus (D \cup OPT)\}$. Delete from G all edges in E_1 . After the deletion, $D, OPT, V \setminus (D \cup OPT)$ are three independent sets. Edges can only cross among them.

Operation 2. Let $E_2 = \{(v, u) \in E \setminus E_1 \mid v \in D, u \in OPT\}$ be the set of remaining edges between D and OPT . For every vertex $v \in D$ (resp. $u \in OPT$), keep those edges in E_v^{OPT} (resp. E_u^D) and delete the other edges of E_2 incident with v (resp. u). Note that for any edge $(v, u) \in E_v^{OPT}$, vertex v is secured by both $v \in D$ and $u \in OPT$. By the definition of G' , vertex v is a witness of edge $(v, u) \in E(G')$. So, $E_v^{OPT} \subseteq E(G')$. Similar argument shows that $E_u^D \subseteq E(G')$.

Operation 3. Let $E_3 = E \setminus (E_1 \cup E_2)$ be the set of edges between $D \cup OPT$ and $V \setminus (D \cup OPT)$. For every vertex $v \in V \setminus (D \cup OPT)$, there exist two vertices $u_1 \in D$ and $u_2 \in OPT$ such that v is secured by both u_1 and u_2 . Keep edges $(v, u_1), (v, u_2)$ and delete the other edges of E_3 incident with v . Then suppress vertex v and shrink edges (v, u_1) and (v, u_2) into one edge (u_1, u_2) . This operation leads to an edge of G' whose existence is witnessed by vertex v . Execute this operation for every vertex $v \in V \setminus (D \cup OPT)$ and delete redundant edges if parallel edges are created.

Note that after the above operations, only those vertices in $D \cup OPT$ are kept in the remaining graph. Furthermore, any edge of G' is witnessed either by one of its ends (which corresponds to Operation 2) or by a vertex $v \in V \setminus (D \cup OPT)$

(which corresponds to Operation 3). So, the above operations results in exactly G' . Since G is a planar graph and the above operations only involve deleting edges and suppressing degree two vertices, the resulting graph is also planar. The lemma is proved. $\square$

Our analysis makes use of a planar separator theorem as follows.

Theorem 2.3 (Frederickson [11]). *For any planar graph $G = (V, E)$ and a parameter $r \geq 1$, there is a partition of V into disjoint union $V = U \cup V_1 \cup \dots \cup V_{\frac{|V|}{r}}$ satisfying the following conditions: there exist three constants c_1, c_2, c_3 such that*

- (i) $|U| \leq c_1 \frac{|V|}{\sqrt{r}}$,
- (ii) $|V_i| \leq c_2 r$ for every $i \in \{1, 2, \dots, \frac{|V|}{r}\}$,
- (iii) $N(V_i) \cap V_j = \emptyset$ for $i \neq j$ and $i, j \in \{1, 2, \dots, \frac{|V|}{r}\}$,
- (iv) $|N(V_i) \cap U| \leq c_3 \sqrt{r}$ for every $i \in \{1, 2, \dots, \frac{|V|}{r}\}$.

The next theorem gives the performance analysis of Algorithm 1.

Theorem 2.4. *Taking $b = c_2 \left(\frac{3(c_1+c_3)}{\varepsilon} \right)^2$, where c_1, c_2, c_3 are the constants in Theorem 2.3, Algorithm 1 is a polynomial-time $(1 + \varepsilon)$ -approximation which runs in time $|V|^{O(1/\varepsilon^2)}$.*

Proof. Apply Theorem 2.3 on the planar graph G' , taking $r = \left(\frac{3(c_1+c_3)}{\varepsilon} \right)^2$, denote the corresponding partition as $V' = U' \cup V'_1 \cup \dots \cup V'_{\frac{|V'|}{r}}$. Let $D_i = D \cap V'_i$ and $OPT_i = OPT \cap V'_i$.

Claim. *For any $i \in \{1, \dots, \frac{|V'|}{r}\}$, $|OPT_i| + |N_{G'}(D_i) \cap U'| \geq |D_i|$.*

Here, we use $N_{G'}$ to emphasize that the neighbor sets are considered in graph G' . A key observation is that

$$(D \setminus D_i) \cup (OPT_i \cup (N_{G'}(D_i) \cap U')) \text{ is also an SDS.} \quad (1)$$

This is because for any vertex v secured by a vertex $u_1 \in D_i$, there is a vertex $u_2 \in OPT$ that also secures v . So, v is a witness of edge $(u_1, u_2) \in E(G')$, and thus $u_2 \in OPT \cap N_{G'}(D_i)$. Because $N_{G'}(D_i) \subseteq (OPT_i \cup (N_{G'}(D_i) \cap U'))$, vertex v is secured by vertex $u_2 \in (OPT_i \cup (N_{G'}(D_i) \cap U'))$. So, property (1) holds.

Because D is a local optimal solution which cannot be improved by switching a vertex set of size at most b , and because $|D_i| \leq |V_i| \leq c_2 r = b$, we have

$$|D_i| \leq |OPT_i \cup (N_{G'}(D_i) \cap U')| = |OPT_i| + |N_{G'}(D_i) \cap U'|.$$

The claim is proved.

By the above claim and Theorem 2.3, it can be estimated that

$$\begin{aligned} |D| &= |U' \cap D| + \sum_{i=1}^{|V'|/r} |D_i| \\ &\leq |U'| + \sum_{i=1}^{|V'|/r} |OPT_i| + \sum_{i=1}^{|V'|/r} |N_{G'}(D_i) \cap U'| \\ &\leq c_1 \frac{|V'|}{\sqrt{r}} + |OPT| + c_3 \sqrt{r} \cdot \frac{|V'|}{r} \\ &= |OPT| + \frac{c_1 + c_3}{\sqrt{r}} (|D| + |OPT|) \\ &\leq |OPT| + \frac{\varepsilon}{3} (|D| + |OPT|). \end{aligned}$$

Rearranging the terms, for $0 < \varepsilon < 1$,

$$|D| \leq \frac{1 + \varepsilon/3}{1 - \varepsilon/3} |OPT| \leq (1 + \varepsilon) |OPT|.$$

As to the running time, we have analyzed it in Lemma 2.1. $\square$

3. A PTAS for MinSDS in growth-bounded graphs

In this section, we design a PTAS for MinSDS in growth-bounded graphs.

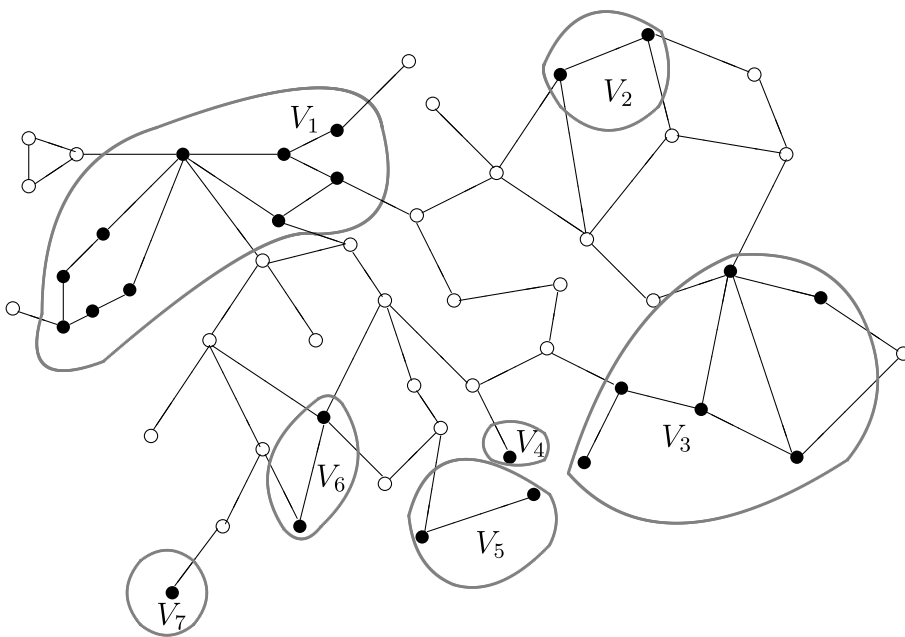


Fig. 1. Example of a 2-separated collection $\{V_1, V_2, V_3, V_4, V_5, V_6, V_7\}$.

3.1. Preliminaries

We first give some notations. Given a graph $G = (V, E)$, a vertex set $I \subseteq V$ is an *independent set* (IS) of G if no vertices in I are adjacent in G . In what follows, $\text{mis}(G)$ is used to denote the *independence number* of G , that is, the size of a maximum IS of graph G . The following observation will be repeatedly used in the proofs.

Observation 3.1. Any maximal IS of graph G is also a DS of G .

For a vertex $v \in V$, denote $N_G^r[v] = \{u \in V \mid d_G(v, u) \leq r\}$ the r -hop closed neighborhood of v in G . For a vertex subset $V' \subseteq V$, denote by $N_G^r[V'] = \bigcup_{v \in V'} N_G^r[v]$ the r -hop closed neighborhood of V' . Let $G[V']$ be the subgraph of G induced by V' , which is a graph on vertex set V' and consists of all those edges of G with both ends belonging to V' . For simplicity of notation, denote $\text{mis}(V') = \text{mis}(G[V'])$. We define a vertex set D an *extended SDS* of V' if $D \subseteq V(G)$ and secures all the vertices in V' , that is, D dominates every vertex of V' and for every $v \in V' \setminus D$, there exists $u \in D$ adjacent to v such that $D \setminus \{u\} \cup \{v\}$ still dominates every vertex of V' . It should be noted that D might contain vertices outside of V' , that is, some vertex in $V \setminus V'$ might be used to dominate some vertex in V' . Note that D can be viewed as a “local” solution because of the observation:

$$\text{if } D \text{ is an extended SDS of } V', \text{ then } D \subseteq N_G[V']. \quad (2)$$

Denote by $D_{\text{ex}}^*(V')$ a minimum extended SDS of V' . It should be noted that

$$|D_{\text{ex}}^*(V')| \leq \gamma_{\text{SDS}}(G[V']) \quad (3)$$

and the inequality might be strict, because $D_{\text{ex}}^*(V')$ has more choices of vertices, other than those in V' , to secure vertices in V' .

Graph G is called *growth-bounded* if there exists a polynomial bounding function $p(r)$ such that $\text{mis}(N_G^r[v]) \leq p(r)$ for any vertex $v \in V(G)$ and any integer $r \geq 0$.

Observation 3.2. For a growth-bounded graph G with bounding function $p(r)$ and any two vertex subsets A, B with $A \subseteq N_G^i[B]$, we have $\text{mis}(A) \leq p(i+1)\text{mis}(B)$.

Proof. Let I_B be a maximum IS of graph $G[B]$. By Observation 3.1, we have $B \subseteq N_G[I_B]$. So $A \subseteq N_G^i[B] \subseteq N_G^{i+1}[I_B] = \bigcup_{v \in I_B} N_G^{i+1}[v]$, and thus $\text{mis}(A) \leq \sum_{v \in I_B} \text{mis}(N_G^{i+1}[v]) \leq p(i+1)|I_B| = p(i+1)\text{mis}(B)$. $\square$

A collection of vertex subsets $\mathcal{C} = \{V_1, V_2, \dots, V_k\}$ is called a *2-separated collection* if $d_G(V_i, V_j) > 2$ for any $i, j \in \{1, 2, \dots, k\}$ and $i \neq j$. Fig. 1 shows an example of a 2-separated collection.

3.2. The Algorithm

Given a growth-bounded graph G with bounding function $p(r)$, we design a PTAS for MinSDS on G as in Algorithm 2. It divides the graph into small subgraphs with bounded radius, computes a minimum extended SDS for each small subgraph and outputs the union of these minimum extended solutions. The division is obtained iteratively in an adaptive manner described as follows. In the algorithm, graph G' is used to record the remaining graph after removing previous subgraphs. In each iteration, the algorithm grows the neighborhood of an arbitrarily picked vertex v in the remaining graph until the first index r_v satisfying the following inequality:

$$\text{mis}(N_{G'}^{r_v+1}[v] \setminus N_{G'}^{r_v}[v]) \leq \varepsilon \cdot \text{mis}(N_{G'}^{r_v}[v]). \quad (4)$$

The vertex set $N_{G'}^{r_v+2}[v]$ forms one part. Compute a minimum extended SDS of $N_{G'}^{r_v+2}[v]$, remove $N_{G'}^{r_v+2}[v]$ from the graph and continue. It should be emphasized that removing $N_{G'}^{r_v+2}[v]$ from G' is only for the purpose of selecting the next part to be processed. Computing $D_{\text{ex}}^*(N_{G'}^{r_v+2}[v])$ is still done in the original graph G , that is, vertices outside of $N_{G'}^{r_v+2}[v]$ might be used to secure vertices in $N_{G'}^{r_v+2}[v]$.

One may ask how to realize the above algorithm efficiently? Because deciding mis (the size of a maximum independent set) and computing a minimum SDS of a graph are generally NP-hard. The key observations are that in a growth-bounded graph, every part with the form $N_{G'}^{r_v+2}[v]$ obtained in the above algorithm has its radius upper bounded by a constant, as shown by Lemma 3.3, and both a maximum IS and a minimum extended SDS for such a part have their sizes upper bounded by a constant, as shown by Corollary 3.5. So, a brute force method can find a maximum IS and a minimum extended SDS of $N_{G'}^{r_v+2}[v]$ in polynomial time. In the following, when the graph G under consideration is clear in the context, we omit the subscript G for simplicity of notation. It should be noted that for a growth-bounded graph, any of its subgraph is also a growth-bounded graph, bounded by a same bounding function.

Algorithm 2 A PTAS for MinSDS in growth bounded graph

Input: A growth-bounded graph $G = (V, E)$ and a constant $\varepsilon > 0$.

Output: A vertex set D which is a $(1 + O(\varepsilon))$ -approximate SDS of G .

```

1:  $D \leftarrow \emptyset$ ;  $G' \leftarrow G$ ;
2: while  $V(G') \neq \emptyset$  do
3:   pick any vertex  $v \in V(G')$ ;  $r_v \leftarrow 0$ ;
4:   while  $\text{mis}(N_{G'}^{r_v+1}[v] \setminus N_{G'}^{r_v}[v]) > \varepsilon \cdot \text{mis}(N_{G'}^{r_v}[v])$  do
5:      $r_v \leftarrow r_v + 1$ ;
6:   end while
7:    $D \leftarrow D \cup D_{\text{ex}}^*(N_{G'}^{r_v+2}[v])$ ;
8:    $G' \leftarrow G' - N_{G'}^{r_v+2}[v]$ ;
9: end while
10: Output  $D$ .
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Lemma 3.3. For a growth-bounded graph G with bounding function $p(r)$ and a constant $\varepsilon > 0$, there is a constant $c = O(1/\varepsilon \cdot \log(1/\varepsilon))$ such that for any vertex v , the first index r_v that satisfies (4) is upper bounded by c .

Proof. Note that such r_v exists, which is upper bounded by the diameter of the graph, because $N^{r+1}[v] \setminus N^r[v] = \emptyset$ if $r \geq$ the diameter. What we need to show is that r_v can be bounded by a constant. The following claim shows that before r_v , the independence number increases exponentially.

Claim. For any $2 \leq i \leq r_v - 1$, $\text{mis}(N^i[v]) > (1 + \varepsilon/p(2)) \cdot \text{mis}(N^{i-2}[v])$.

Note that $N^{i+1}[v] \setminus N^i[v] \subseteq N(N^i[v] \setminus N^{i-1}[v])$. So by Observation 3.2, we have

$$\text{mis}(N^{i+1}[v] \setminus N^i[v]) \leq p(2)\text{mis}(N^i[v] \setminus N^{i-1}[v]). \quad (5)$$

Since r_v is the first index satisfying (4), for any $i \leq r_v - 1$, we have $\text{mis}(N^{i+1}[v] \setminus N^i[v]) > \varepsilon \cdot \text{mis}(N^i[v])$. Combining this with (5),

$$\begin{aligned} \text{mis}(N^i[v] \setminus N^{i-1}[v]) &\geq 1/p(2) \cdot \text{mis}(N^{i+1}[v] \setminus N^i[v]) \\ &> \varepsilon/p(2) \cdot \text{mis}(N^i[v]) \\ &\geq \varepsilon/p(2) \cdot \text{mis}(N^{i-2}[v]). \end{aligned} \quad (6)$$

Note that $N^i[v] \setminus N^{i-1}[v]$ and $N^{i-2}[v]$ are 1-hop separated and thus the union of an IS in $N^i[v] \setminus N^{i-1}[v]$ and an IS in $N^{i-2}[v]$ is still independent. Combining this with (6),

$$\begin{aligned} \text{mis}(N^i[v]) &\geq \text{mis}(N^i[v] \setminus N^{i-1}[v]) + \text{mis}(N^{i-2}[v]) \\ &> (1 + \varepsilon/p(2)) \cdot \text{mis}(N^{i-2}[v]). \end{aligned} \quad (7)$$

Suppose r_v is an odd number (the case that r_v is an even number can be argued similarly), recursively using (7), we have

$$\text{mis}(N^{r_v-1}[v]) > (1 + \varepsilon/p(2))^{\frac{r_v-1}{2}} \cdot \text{mis}(N^0[v]) = (1 + \varepsilon/p(2))^{\frac{r_v-1}{2}}.$$

By the definition of growth-bounded graph,

$$\text{mis}(N^{r_v-1}[v]) \leq p(r_v - 1).$$

Combining the above two inequalities,

$$r_v \leq 1 + 2 \log_{1+\varepsilon/p(2)} p(r_v), \quad (8)$$

which is equivalent to

$$\frac{\log(1 + \varepsilon/p(2))}{2} \leq \frac{\log p(r_v)}{r_v - 1}.$$

Because $p(r_v)$ is a polynomial bounding function with respect to r_v , the right hand side of the above inequality is $O(\log r_v/r_v)$. Using Taylor' expansion $\ln(1+x) = x - \frac{x^2}{2} + \dots$, the left hand side of the above inequality is $O(\varepsilon)$ (note that $p(2)$ is a constant). So

$$\frac{r_v}{\log r_v} \leq O\left(\frac{1}{\varepsilon}\right),$$

which implies that $r_v = O(1/\varepsilon \cdot \log(1/\varepsilon))$. $\square$

The next lemma reveals some relations between the size of a minimum extended SDS and the size of a maximum IS of a growth-bounded graph.

Lemma 3.4. For a growth-bounded graph G with bounding function $p(r)$ and a vertex subset $V' \subseteq V$, we have $|D_{\text{ex}}^*(V')| \leq 2\text{mis}(V')$ and $\text{mis}(V') \leq p(1)|D_{\text{ex}}^*(V')|$.

Proof. Suppose I is a maximum IS of $G[V']$ and I' is a maximum IS of $G[V' - I]$. Note that by Observation 3.1, any vertex $v \in V' \setminus (I \cup I')$ is adjacent to some vertex $u_v \in I$ and some vertex $u'_v \in I'$, which implies that both $I \cup I'$ and $(I \cup I') \setminus \{u_v\} \cup \{v\}$ are DSs of $G[V']$. So, $I \cup I'$ is an SDS of $G[V']$. It follows that $\gamma_{\text{SDS}}(G[V']) \leq |I \cup I'| \leq |I| + |I'| \leq 2|I| = 2\text{mis}(V')$. Combining this with (3), the first inequality is proved.

To prove the second inequality, note that $V' \subseteq N_G[D_{\text{ex}}^*(V')] = \bigcup_{v \in D_{\text{ex}}^*(V')} N_G[v]$. So, $\text{mis}(V') \leq \sum_{v \in D_{\text{ex}}^*(V')} \text{mis}(N_G[v]) \leq p(1)|D_{\text{ex}}^*(V')|$. $\square$

Combining the definition of growth-bounded graph with Lemmas 3.3 and 3.4, we have the following corollary.

Corollary 3.5. Suppose G is a growth-bounded graph with bounding function $p(r)$. For any vertex v picked in the algorithm, all mis-values that are needed by the execution of the algorithm is upper bounded by $p(c+1)$ and the minimum extended SDS has size $|D_{\text{ex}}^*(N_{G'}^{r_v+2}[v])| \leq 2\text{mis}(N_{G'}^{r_v+2}[v]) \leq 2p(c+2)$, where c is the constant guaranteed by Lemma 3.3. Hence the running time of Algorithm 2 is $O(n^{2p(c+2)})$.

3.3. Approximation ratio

Before analyzing the approximation ratio, we first prove the following lemma, which relates the extended secure domination numbers of $N_{G'}^{r_v}[v]$ and $N_{G'}^{r_v+2}[v]$.

Lemma 3.6. Suppose G is a growth-bounded graph with bounding function $p(r)$. For any vertex v picked in the algorithm, we have $|D_{\text{ex}}^*(N_{G'}^{r_v+2}[v])| \leq (1 + 2p(2)p(1)\varepsilon)|D_{\text{ex}}^*(N_{G'}^{r_v}[v])|$.

Proof. Since $N_{G'}^{r_v+2}[v] \setminus N_{G'}^{r_v}[v] \subseteq N_{G'}[N_{G'}^{r_v+1}[v] \setminus N_{G'}^{r_v}[v]]$, by Observation 3.2, we have

$$\text{mis}(N_{G'}^{r_v+2}[v] \setminus N_{G'}^{r_v}[v]) \leq p(2) \cdot \text{mis}(N_{G'}^{r_v+1}[v] \setminus N_{G'}^{r_v}[v]).$$

Since r_v satisfies (4), we have

$$\text{mis}(N_{G'}^{r_v+2}[v] \setminus N_{G'}^{r_v}[v]) \leq p(2)\varepsilon \cdot \text{mis}(N_{G'}^{r_v}[v]).$$

Then, by [Lemma 3.4](#),

$$\begin{aligned} & |D_{ex}^*(N_{G'}^{r_v+2}[v] \setminus N_{G'}^{r_v}[v])| \\ & \leq 2\text{mis}(N_{G'}^{r_v+2}[v] \setminus N_{G'}^{r_v}[v]) \\ & \leq 2p(2)\varepsilon \cdot \text{mis}(N_{G'}^{r_v}[v]) \\ & \leq 2p(2)p(1)\varepsilon |D_{ex}^*(N_{G'}^{r_v}[v])|. \end{aligned}$$

Since $D_{ex}^*(N_{G'}^{r_v+2}[v] \setminus N_{G'}^{r_v}[v])$ is a vertex set that secures vertices in $N_{G'}^{r_v+2}[v] \setminus N_{G'}^{r_v}[v]$, and $D_{ex}^*(N_{G'}^{r_v}[v])$ secures vertices in $N_{G'}^{r_v}[v]$, their union secures all vertices in $N_{G'}^{r_v+2}[v]$. So,

$$\begin{aligned} & |D_{ex}^*(N_{G'}^{r_v+2}[v])| \\ & \leq |D_{ex}^*(N_{G'}^{r_v}[v])| + |D_{ex}^*(N_{G'}^{r_v+2}[v] \setminus N_{G'}^{r_v}[v])| \\ & \leq (1 + 2p(2)p(1)\varepsilon) |D_{ex}^*(N_{G'}^{r_v}[v])|. \end{aligned}$$

The lemma is proved. $\square$

We now prove that [Algorithm 2](#) is a PTAS for MinSDS on growth-bounded graphs.

Theorem 3.7. For a growth-bounded graph G with bounding function $p(r)$, [Algorithm 2](#) computes in time $O(n^{2p(c+2)})$ an SDS of G that approximates the optimal solution within factor $1 + \varepsilon_1$, where $\varepsilon_1 = 1 + 2p(2)p(1)\varepsilon$ and $c = O(1/\varepsilon \cdot \log(1/\varepsilon))$.

Proof. Suppose the vertices picked in [Algorithm 2](#) are sequentially $v_1, \dots, v_k$. For $i = 1, \dots, k$, denote the remaining graph G' in the iteration when v_i is picked as G_i , and let $V_i = N_{G_i}^{r_{v_i}}[v_i]$. Denote by OPT an optimal solution to the MinSDS problem on G . The update method of G' in line 8 of [Algorithm 2](#) implies that $V_1, \dots, V_k$ form a 2-separated collection of G . So, $N_G[V_i] \cap N_G[V_j] = \emptyset$ and thus

$$|OPT| \geq \sum_{i=1}^k |OPT \cap N_G[V_i]|.$$

Because securing all the vertices in V_i only involves vertices in $N_G[V_i]$, we see that $OPT \cap N_G[V_i]$ is an extended SDS of V_i . So

$$|D_{ex}^*(V_i)| \leq |OPT \cap N_G[V_i]|$$

It follows that

$$|OPT| \geq \sum_{i=1}^k |D_{ex}^*(V_i)|. \quad (9)$$

The vertex set D output by [Algorithm 2](#) has $D = \bigcup_{i=1}^k D_{ex}^*(N_{G_i}^{r_{v_i}+2}[v_i])$. So by [Lemma 3.6](#) and inequality (9), the size of D satisfies

$$\begin{aligned} |D| & \leq \sum_{i=1}^k |D_{ex}^*(N_{G_i}^{r_{v_i}+2}[v_i])| \\ & \leq (1 + 2p(2)p(1)\varepsilon) \sum_{i=1}^k |D_{ex}^*(N_{G_i}^{r_{v_i}}[v_i])| \\ & = (1 + 2p(2)p(1)\varepsilon) \sum_{i=1}^k |D_{ex}^*(V_i)| \\ & \leq (1 + 2p(2)p(1)\varepsilon) |OPT|. \end{aligned}$$

The approximation ratio is proved.

The time complexity follows from [Corollary 3.5](#) and [Lemma 3.3](#). $\square$

4. Conclusions

In this paper, we design PTASs for MinSDS on planar graphs and growth-bounded graphs, using a local search method and a divide-and-conquer method, respectively. For the local search algorithm, the crucial step is how to construct a planar graph satisfying a locality condition. For the divide-and-conquer algorithm, the crucial steps are how to partition the graph into subgraphs with constant sizes (this is done through an adaptively growing method) and how to relate

the optimal solution restricted onto one part and the minimum extended solution (this is done through a concept called 2-separated collection). As far as we know, there is no approximation algorithm for MinSDS on general graphs. How to design approximation algorithms for a general graph remains to be further explored.

Data availability

No data was used for the research described in the article.

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